

# NCTU-EE IC Design LAB – Fall2024

## Final Project: Innovus - From RTL to GDSII

Due to the main goal of this course is to teach students on the front-end design methodology, the APR of final project will be modified to pin design rather than pad design so it's easier.

### 1. Data Preparation

1. Complete rtl simulation, synthesis, and gate level simulation.
2. **cd ./05\_APR/ write and check your CHIP.sdc file.**  
(Hint: you may modify the file generated during synthesize in ../02\_SYN/Netlist folder, and carefully check the constraints following the check\_list.pdf document.)
3. **cp ISP\_SYN.v CHIP\_SYN.v** and change the **module name** to **CHIP**. (You can use command 02\_COPY under 05\_APR to do so.)  
  
(Because the design does not require any pad connection, you won't need to write your own shell file, nor do you need to execute ./00\_combine.)
4. **You don't need to write the CHIP.io file when you run Innovus at the first time.**  
(The import design step should use CHIP\_SYN.v and leave the CHIP.io part empty.)  
  
Notice that TA has provided 05\_APR/mmmc.view for you, please do not create mmmc.view by yourself, or you may fail in demo.
5. Open file: 05\_APR/cmd/run\_apr.cmd  
This file sets up the initial design environment. You can use these commands as an alternative to performing actions step-by-step via the GUI. However, the GUI option remains available if you find it more convenient.

```
# source cmd/floorPlan.cmd
# source cmd/powerRing.cmd
# source cmd/powerStripe.cmd
# source cmd/addIOFiller.cmd
# source cmd/place.cmd
# source cmd/ccopt.cmd
# source cmd/postCTSTiming.cmd
# source cmd/nanoRoute.cmd
# source cmd/postRouteTiming.cmd
# source cmd/postRouteVerify.cmd
# source cmd/addFiller.cmd
# source cmd/signOff.cmd
# source cmd/streamOut.cmd
```

Notice that please add your memory LEF file PATH at the end of lefFile variable.

For example:

add **../04\_MEM/XXXX.lef** beyond the \$ProcessRoot/lef/BONDPAD.lef

```
# =====
# Project:  iclab APR Flow
# File:      run_apr.cmd
# Author:    Lai, Lin-Hung @ Si2 Lab
# Date:      2023.07.25
# =====
##### Parameter setting #####
#####
set ProcessRoot "/RAID2/COURSE/BackUp/2023_Spring/iclab/iclabta01/UMC018_CBDK/CIC/SOCE/"
set MemoryRoot  "../Memory/ftclib_200901.2.1/EXE/"
set NUM_OF_CPU 48
set mmmcFile "mmmc.view"
set lefFile "
    $ProcessRoot/lef/header6_V55_20ka_cic.lef
    $ProcessRoot/lef/fsa0m_a_generic_core.lef
    $ProcessRoot/lef/FSA0M_A_GENERIC_CORE_ANT_V55.lef
    $ProcessRoot/lef/fsa0m_a_t33_generic_io.lef
    $ProcessRoot/lef/FSA0M_A_T33_GENERIC_IO_ANT_V55.lef
    $ProcessRoot/lef/BONDPAD.lef
"
set topDesign "CHIP"
set verilogFile "./CHIP_SYN.v"
set ioFile      "./CHIP.io"
set pwrNet      "VCC"
set gndNet      "GND"
```

6. Launch innovus by the following command:

```
innovus -cpus 8 -init ./cmd/run_apr.cmd -log ./log/run_apr.log -cmd
./log/run_apr.cmdlog
```

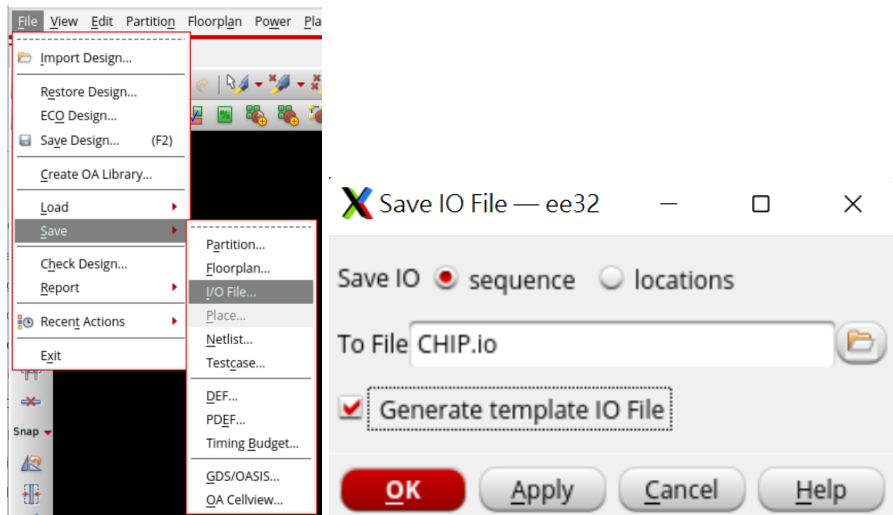
or

```
make apr_init
```

you may find this error due to no CHIP.io provided, thus you can do step 7 to auto generate this io file.

```
** Memory Usage v#1 (Current mem = 982.402M, initial mem = 292.375M) **
**ERROR: (IMPSYUTIL-96): Cannot open (for read) FE IO file file: './CHIP.io'. The reason is: No such file or directory
Type 'man IMPSYUTIL-96' for more detail.
**WARN: (IMPFP-110): Failed to open file './CHIP.io' for reading.
**ERROR: (IMPSE-110): File './cmd/run_apr.cmd' line 434: errors out.
innovus 1> █
```

7. Upper left corner: FILE -> SAVE -> IO FILE ->  
sequence check & generate template IO file check  
change the file name to CHIP.io  
and click OK.



Then exit, restart innovus again. You may see the pin shape below.



8. You may change the orders in CHIP.io file if you wish to modify your pin orders.

```
#####
# Generated by: Cadence Innovus 20.15-s105_1
# OS: Linux x86_64(Host ID ee32)
# Generated on: Fri Dec 22 15:47:26 2023
# Design: CHIP
# Command: saveIoFile -byOrder -temp CHIP.io
#####

(globals
  version = 3
  io_order = default
)
(iopin
  (top
    (pin name="clk" layer=1 place_status=unplaced )
    (pin name="awaddr_m_inf[31]" layer=1 place_status=unplaced )
    (pin name="awaddr_m_inf[24]" layer=1 place_status=unplaced )
    (pin name="awaddr_m_inf[17]" layer=1 place_status=unplaced )
    (pin name="awaddr_m_inf[10]" layer=1 place_status=unplaced )
    (pin name="awaddr_m_inf[3]" layer=1 place_status=unplaced )
    (pin name="awburst_m_inf[1]" layer=1 place_status=unplaced )
    (pin name="awlen_m_inf[1]" layer=1 place_status=unplaced )
    (pin name="wdata_m_inf[12]" layer=1 place_status=unplaced )
    (pin name="wdata_m_inf[5]" layer=1 place_status=unplaced )
    (pin name="wvalid_m_inf[0]" layer=1 place_status=unplaced )
    (pin name="bresp_m_inf[0]" layer=1 place_status=unplaced )
    (pin name="arid_m_inf[3]" layer=1 place_status=unplaced )
    (pin name="araddr_m_inf[60]" layer=1 place_status=unplaced )
    (pin name="araddr_m_inf[53]" layer=1 place_status=unplaced )
  )
)
```

9. After that, most of the steps are the same as Lab11 and Lab12. The only difference is that there are no pads, so you should only add block pin in **(Connect Core Power Pin)** and step **(Add PAD Filler)**. Please refer to the 05\_APR/cmd/Final\_Project.cmd

## 2. SDC FILE

### SDC file CHIP.sdc

This file is generated from 02\_SYN.

You may change this to your target cycle time by yourself.

Normally, this may be equal or larger than 02\_SYN due to wiring delay.

```
create_clock [get_ports clk] -period 20 -waveform {0 10}
set_clock_uncertainty 0.1 [get_clocks clk]
set_input_delay -clock clk 0 [get_ports clk]
```

Before CTS stage, you may no need to comment out the set\_clock\_uncertainty & set\_clock\_transition instruction.

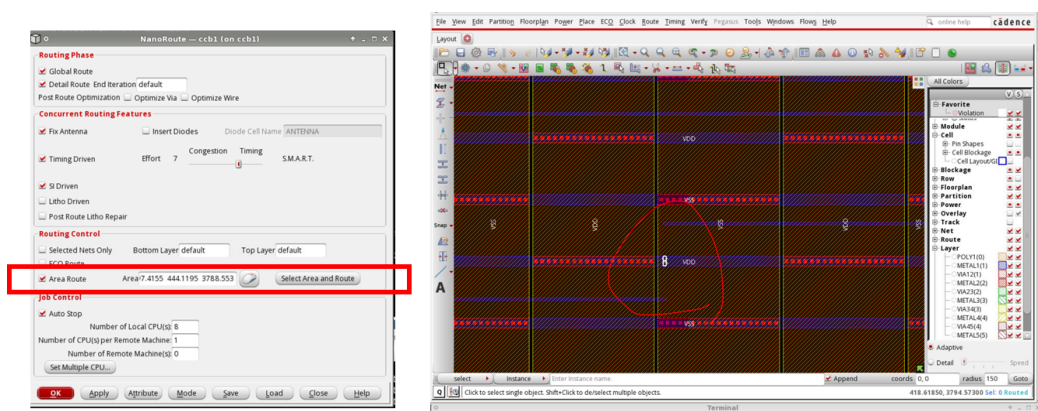
### SDC file CHIP\_cts.sdc

The only difference of CHIP\_cts.sdc is that you need to comment out the set\_clock\_uncertainty & set\_clock\_transition instruction.

Then do the CTS stage.

## 3. Short issue

Delete the short metal line, then use area nano route (Area Route--> Select Area and Route) to re-route that path. (sometimes may need to delete a longer metal line so the new routed result won't have short again)



## 4. Check Area

type > **summaryReport -noHtml -outfile summaryReport.rpt**

then you will see the CORE area.